

“Automated Quality Control System using Machine Learning”



*A Project Report Submitted to
Rajiv Gandhi Proudtyogiki
Vishwavidyalaya, Bhopal towards the
partial fulfillment of the degree of
Bachelor of Technology in Mechanical
Engineering*

2022-2026

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A Govt. Aided Autonomous Institute, Affiliated to RGPV, Bhopal
DEPARTMENT OF MECHANICAL ENGINEERING

2022 - 26



RECOMMENDATION

We are pleased to recommend that the project work entitled “**Automated Quality Control System using Machine Learning**” submitted by **Romil Dubey, Hiya Sethi, Sandhya Gupta, Yashveer Singh and Prateek Chourasiya** may be accepted in partial fulfillment of the degree of **Bachelor of Technology in Mechanical Engineering** of Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (M.P.) during the session **2022-26**.

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CERTIFICATE

This is to certify that the Project entitled
**“Automated Quality Control System using
Machine Learning”** submitted by **Romil
Dubey, Hiya Sethi, Sandhya Gupta,
Yashveer Singh and Prateek Chourasiya** is
accepted in partial fulfillment of the degree of
**Bachelor of Technology in Mechanical
Engineering** of Rajiv Gandhi Proudyogiki
Vishwavidyalaya, Bhopal during the **2022-26**.

Internal Examiner

Date:

External Examiner

Date:

DECLARATION

We, Romil Dubey, Hiya Sethi, Sandhya Gupta, Yashveer Singh and Prateek Chourasiya declare that the Project entitled **“Automated Quality Control System using Machine Learning”** is our own work conducted under the supervision of **Dr. R.K. Porwal (Professor, MED, SGSITS)** and co-supervision of **Dr. Sumer Singh Patel (Assistant Professor, MED, SGSITS)**.

We affirm that this work is entirely original and has not been copied from any source. All sources consulted during the research process have been duly acknowledged and cited in the bibliography.

We further declare that to the best of our knowledge, this project work does not contain any part of any work which has been submitted for the award of any degree or any other work either in this University or in any other University/ websites without proper citation.

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Date: May 6, 2026.

ABSTRACT

The modern agricultural and food processing industry relies heavily on consistent quality control to ensure consumer safety, minimize waste, and maximize economic returns. However, a significant technological gap exists for small-to-medium-sized enterprises and local distributors who cannot afford the high capital expenditure required for industrial-scale automated sorting systems. These operators typically rely on manual labor, which is inherently subjective, prone to fatigue, and serves as a major bottleneck in the supply chain. This project presents the design and development of an **Automated Quality Control System using Machine Learning**, engineered specifically to provide a cost-effective, high-throughput solution for sorting fruits based on visual and structural defects.

The system integrates advanced computer vision, digital fabrication, and embedded hardware to create a functional prototype capable of real-time autonomous decision-making. The software core utilizes a **Two-Stage Cascade Architecture** powered by the **YOLOv8** framework. The first stage employs YOLOv8n for real-time object localization, while the second stage uses a custom-trained YOLOv8n-cls model for granular classification. To enhance robustness beyond simple color detection, the model incorporates **texture-based feature extraction** using Gray-Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP) to identify organic degradation like bruising and mold. The machine learning pipeline, deployed via a Python-based Flask server, achieves a Top-1 Accuracy of **98.7%** for banana ripeness and an overall multi-fruit accuracy of **94%** for bananas, apples, and oranges. The mechanical architecture consists of a modular **ladder-frame conveyor system** designed in **Autodesk Fusion 360** and fabricated using **PLA 3D printing**.

The transport zone is powered by dual 12V DC gear motors, while the "Decision Execution Unit" utilizes an **Arduino Mega 2560** to control a high-torque **MG995 servo motor**. A specialized 3D-printed deflection lever (Extender) intercepts "Rotten" payloads based on serial signals transmitted from the ML model. To ensure system stability, a dedicated **5V 3A external power supply** with a common ground configuration is used to isolate the high-current servo loads from the logic circuitry. The system achieved a total inference-to-actuation latency of **<30ms**, supporting high-speed conveyor operation. Through value engineering, the prototype cost was successfully optimized to under **₹5,000**, demonstrating that industrial-grade quality control can be achieved at a fraction of the cost of traditional systems.

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Major Project Phase - 1

Chapter 1 - Introduction

1.1 Background and Motivation:

Quality control is a critical pillar of the modern agricultural and food processing industry, ensuring consumer safety, reducing waste, and maximizing profitability. While large-scale agricultural corporations employ sophisticated, high-cost sorting systems, a significant technological and economic gap exists for small to medium-sized farms and local distributors. These smaller enterprises often rely on manual labour for sorting, a process that is slow, inconsistent, and subject to human fatigue and error. This project is motivated by the need to bridge this gap by leveraging accessible, open-source technologies. By integrating affordable microcontrollers like Arduino, simple mechanical actuators like servo motors, and powerful open-source software like OpenCV for computer vision, we aim to demonstrate a viable and cost-effective model for automated quality control in smaller-scale operations.

1.2 Problem Statement:

The primary problem is the lack of affordable, efficient, and data-driven automated sorting solutions for small-scale fruit processing. Manual sorting of fruits is labor-intensive, subjective, and a significant bottleneck that can lead to inconsistent product quality and increased spoilage. Existing industrial automation systems are often prohibitively expensive and too complex for smaller operators. Therefore, this project aims to solve the technical challenge of designing and building a cost-effective automated system that can autonomously sort fruits from a conveyor belt based on visual quality (spoiled vs. good), using a machine learning model for detection and a simple mechanical system for separation.

1.3 Objectives:

The primary objective of this project is to design, build, and demonstrate an automated sorting system capable of differentiating between good and spoiled fruits. The secondary objectives are:

1. To develop a computer vision model using Python and OpenCV to accurately detect and classify fruits based on visual defects.
2. To construct a small-scale conveyor belt system integrated with a camera for real-time image capture.
3. To implement a sorting mechanism using an Arduino microcontroller and a servo motor that physically separates fruits based on the classification from the vision model.

1.4 Scope of the Project:

The scope of this project includes the complete design-to-demonstration lifecycle of the fruit sorting system. This encompasses the mechanical design of the conveyor and sorting apparatus; the design of the electronic control circuit for the motors and sensors; the development of the computer vision software for fruit classification; and the final integration and testing of the complete system. The project will focus on sorting one or two specific types of fruit (e.g., apples, oranges) based on visible defects like rot, significant blemishes, and discoloration in a controlled lighting environment.

Chapter 2 - Literature Review

2.1 Automated Sorting Mechanisms in Agriculture

The use of automated sorting mechanisms is well-documented in agricultural processing. High-end industrial systems often use pneumatic ejectors or high-speed robotic arms for their precision and speed, but these come at a significant cost. In academic and small-scale applications, there is a strong focus on cost-effective alternatives. Research has demonstrated the viability of simple mechanical actuators, such as servo-controlled gates, flippers, or pushers, for effective sorting tasks. These systems, while simpler, provide a robust and affordable solution for separating items on a conveyor, which forms the basis for our mechanical design.

2.2 Conveyor Systems for Material Handling

Conveyor systems are fundamental to automation in the food industry. They provide a consistent and controlled method for transporting items through various processing stages, including inspection and sorting. The design of these systems often prioritizes gentle handling to prevent bruising or damage to the produce. The integration of a variable-speed conveyor with a vision system is a common architecture, allowing for synchronization between item detection and ejection. This established approach validates our use of a small-scale conveyor as the core transportation element in our system.

2.3 Computer Vision for Fruit Quality Inspection

Computer vision is the primary non-contact sensing modality for automated quality control in agriculture. Traditional techniques often rely on color thresholding in spaces like HSV or simple shape analysis. While effective for basic grading, these methods struggle with the natural variability and complex defects found in fruits. A more robust approach involves the YOLO architecture. By orchestrating multiple YOLO instances, our system separates object detection from fine-grained defect classification, identifying rot and bruises with high precision. The real-time inference speed of these models makes this specific architecture a superior choice for our project's scope.

2.4 Microcontrollers in Automated Control Systems

The control architecture of small-scale automation systems often revolves around low-cost microcontrollers. Platforms like Arduino are ubiquitous in both academic research and DIY projects due to their simplicity, extensive community support, and ease of interfacing with a wide range of sensors and actuators. In the context of a sorting system, the microcontroller acts as the central hub, receiving classification signals from the computer vision system and translating them into precise actions for the servo motor to execute the sort. The reliability and real-time processing capabilities of these devices are well-proven, making them an ideal choice for controlling the physical components of our system.

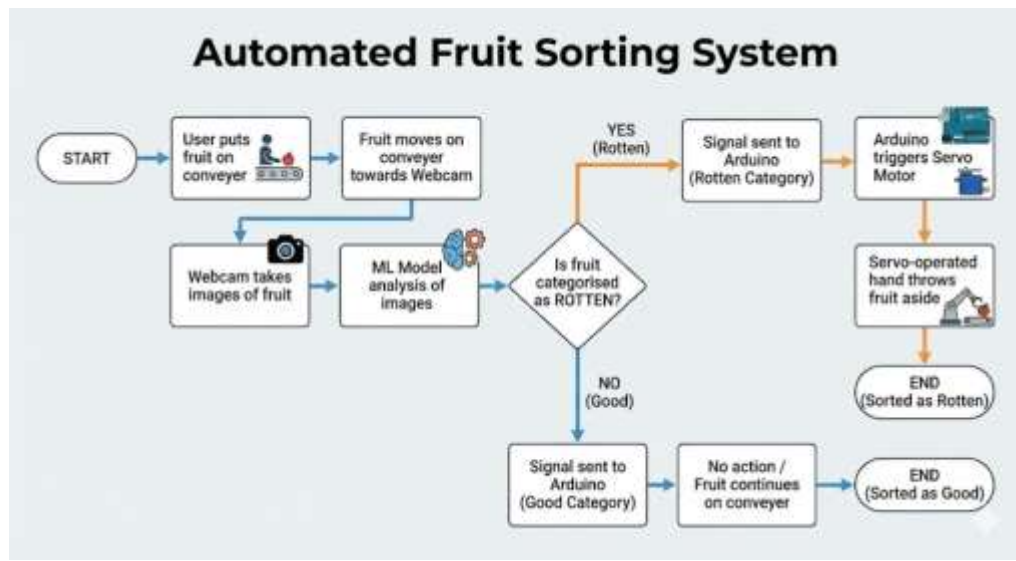


Figure 1: Flowchart of the Automated Quality Control System using Machine Learning

Chapter 3: Methodology and Timeline

3.1 System Architecture Design

The system is designed using a hierarchical Master-Slave control architecture. This model decouples high-level processing from real-time control, enhancing system stability and modularity.

- **The Master (Laptop):** Acts as the central brain. It runs the main Python script responsible for perceiving the environment (via the camera), making intelligent decisions, and delegating tasks to the slave.
- **The Slave (Arduino Mega):** Acts as the muscle. Its sole purpose is to execute commands received from the master as precisely as possible

3.2 Project Timeline (32 Weeks)

The project is structured over two semesters (approximately 32 weeks) to allow for thorough research, design, implementation, and testing.

Table 1: Project Timeline (32 Weeks)

Phase	Weeks	Key Tasks	Major Deliverables
I	1-4	Research and Feasibility (Completed)	Finalized component list (Camera, Arduino, Servo, etc.), concepts for conveyor & sorter, Feasibility Report.
	5-8	Full System Integration & Core Programming (Completed)	Mechanical and electrical systems integrated, Core software linking OpenCV to Arduino control.
	9-12	ML Model Training & Refinement (Completed)	Functional and trained OpenCV model (>90% accuracy), Implemented error-handling logic for detection.
	13-16	Sub-System Testing & Mid-Term Review (Completed)	Independent tests of conveyor, servo sorter, and initial vision model. Mid-

			term presentation.
II	17-20	Detailed Design & Procurement (Completed)	Multi-fruit dataset acquisition (Apple/Orange) and Fabrication of Mega 2560 Shield & 5V Power Rails.
	21-24	Fabrication & Sub-System Assembly (Completed)	3D printed /fabricated parts, Assembled conveyor and sorting mechanism, Mounted camera.
	25-28	Final Testing & Performance Optimization (Completed)	Full system endurance tests, Final system calibration.
	29-32	Final Documentation & Project Demonstration (Ongoing)	Final Project Report, Final Presentation and Demonstrati on.

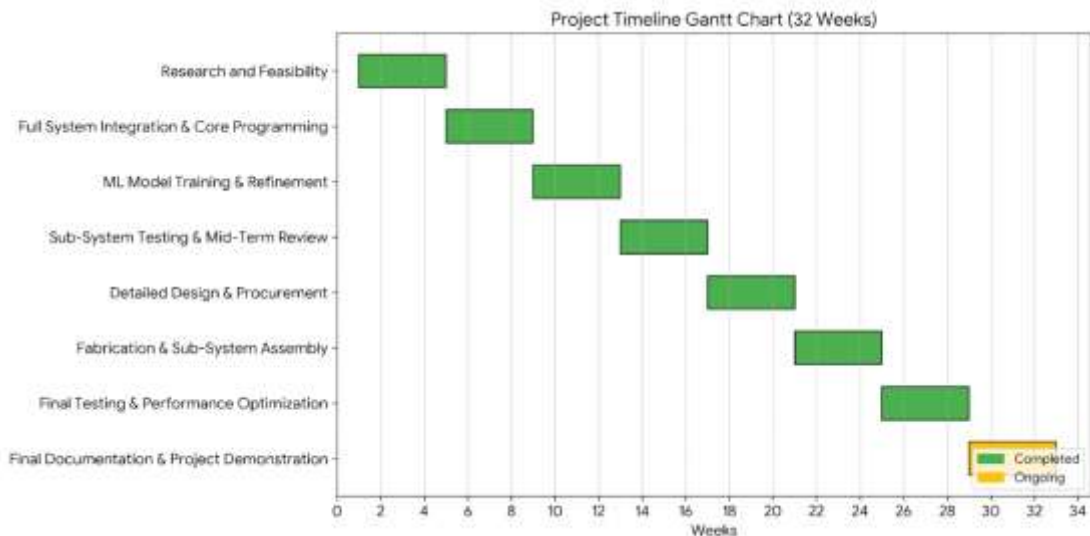


Figure 2: Gantt Chart of the Project Timeline

Chapter 4: Preliminary Work and Feasibility Study

This chapter details the foundational work completed in the first four weeks, which has validated the core concepts of the project and confirmed the viability of our chosen approach.

4.1 Component Selection:

Based on our research, a list of primary components has been finalized. The selection prioritizes a balance of cost, performance, and ease of integration.

Table 2: Mechanical Components Selection

Component	Material/Model	Purpose
Conveyor Frame & Supports	PLA	Provides a lightweight and rigid structural frame for the entire system.
Conveyor Belt	Rubber	Transports fruits gently through the inspection and sorting zones.
Drive & Idler Pulleys	PLA	Transfer motion from the DC motor to the belt and maintain tension.
Sorting Lever/Gate	PLA	Actuated by a servo motor to physically separate good and faulty fruits.

Table 3: Specifications of Electronic Components

Component	Material/Model	Purpose
Processor	Laptop	Processing for real-time fruit detection using the OpenCV model.
Microcontroller	Arduino Mega 2560	Controls the conveyor motor and servo on commands from the processor.

Conveyor Motor	DC Gear Motor	Drives the conveyor belt to transport fruit.
Sorting Servo	MG995 Metal Gear High-Torque Servo Motor.	Actuates the sorting lever/gate to separate good and faulty fruit.
Vision Sensor	USB Webcam	Captures video feed for the vision system.

Chapter 5: Initial Computer Vision Model Training

1. Project Overview and Objectives

The core objective of this module is to replace subjective manual inspection with a deterministic, quantitative computer vision system. The system is engineered to function in a high-throughput agricultural environment, specifically targeting the classification of bananas into four distinct biological stages.

Primary Technical Objectives:

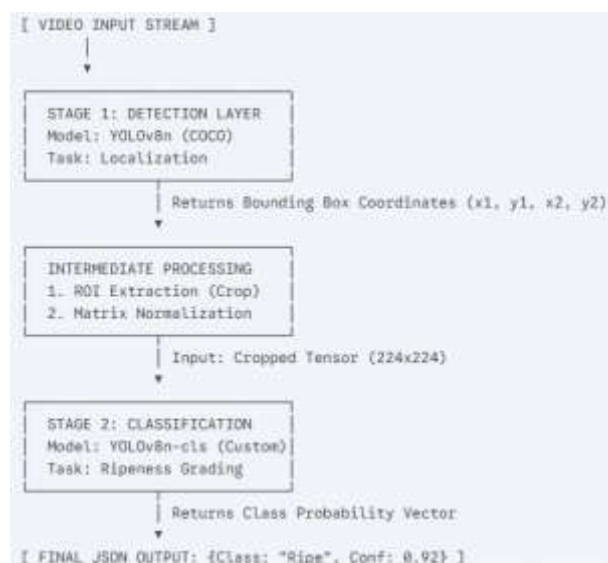
- **Granular Classification:** Distinguish between four specific ripeness stages:
 - **Unripe:** Characterized by dominant green pixel values.
 - **Ripe:** Characterized by dominant yellow pixel values.
 - **Overripe:** Yellow surface with brown oxidization spots.
 - **Rotten:** Structural degradation and dominant black/brown coloration.
- **Real-Time Inference:** Achieve a processing latency of $<30\text{ms}$ per frame to enable live conveyor belt monitoring.
- **Decoupled Architecture:** Utilize a multi-stage inference pipeline to separate object localization (where is the fruit?) from feature classification (what is the quality?).

2. System Architecture and Technology

Unlike standard monolithic models, this project employs a **Cascade Architecture**. This design involves orchestrating two separate instances of the YOLO (You Only Look Once) architecture in a serial pipeline.

2.1 Pipeline Logic Flow

The data flows sequentially through a localization phase and a classification phase as detailed below:



2.2 Component Specifications

The system leverages the Ultralytics YOLOv8 framework, chosen for its state-of-the-art balance between Mean Average Precision (mAP) and inference speed.

Table 4: Table showing the use of multiple instances of YOLO

Component	Model Version	Weights	Responsibility	Configuration
Detector	YOLOv8n (Nano)	yolov8n.pt (Pre-trained on COCO)	Identify generic objects and generate Regions of Interest (ROI).	Confidence Thresh: 0.20
Classifier	YOLOv8n-cls (Nano)	best.pt (Custom Trained)	Analyze pixel texture and color within the ROI to determine grade.	Image Size: 224 × 224
Backend	PyTorch	v2.5.1 + cu121	Tensor operations and GPU acceleration.	CUDA Core optimization

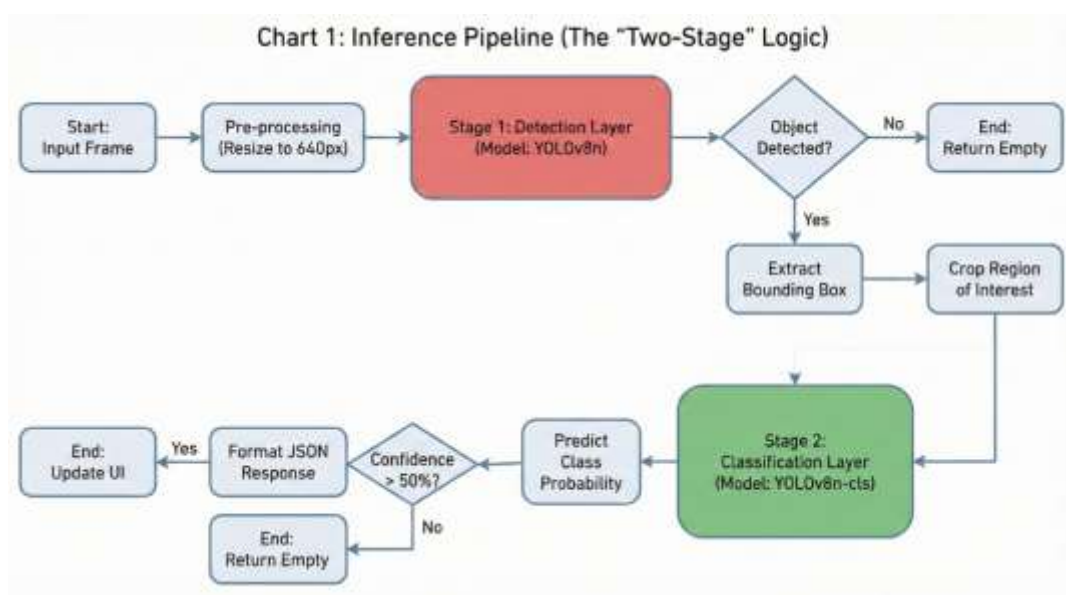


Figure 3: Inference Pipeline of the two-stage logic

3. Data Acquisition and Modelling

3.1 Dataset Composition

The model training relied on a curated dataset of **12,916 images**. To ensure the model is robust against environmental variance, the dataset includes variations in lighting, background noise, and fruit orientation.

Table 5: Table showing multiple fruit class and their sample counts

Ripeness Class	Visual Characteristics	Training Sample Count	Validation Sample Count	Total Samples
Rotten	High surface oxidization, black patches	4,020	388	4,408
Ripe	Uniform yellow hue, minimal spotting	3,522	339	3,861
Overripe	Yellow hue with "sugar spots" (brown dots)	2,349	229	2,578
Unripe	Uniform green hue, rigid structure	1,902	167	2,069
TOTAL		11,793	1,123	12,916

3.2 Training Strategy

The classification model (YOLOv8n-cls) was trained using Transfer Learning, initializing weights from ImageNet and fine-tuning on our custom dataset.

- **Hardware:** NVIDIA RTX 3050 Laptop GPU (4GB VRAM).
- **Optimizer:** Stochastic Gradient Descent (SGD) with momentum.
- **Early Stopping:** A patience mechanism was implemented to halt training if validation loss did not improve for **20 consecutive epochs**. This occurred at Epoch 32, preventing overfitting and saving computational resources.
- **Augmentation pipeline:** Random flip, scale, and color jitter were applied during training to improve generalization.

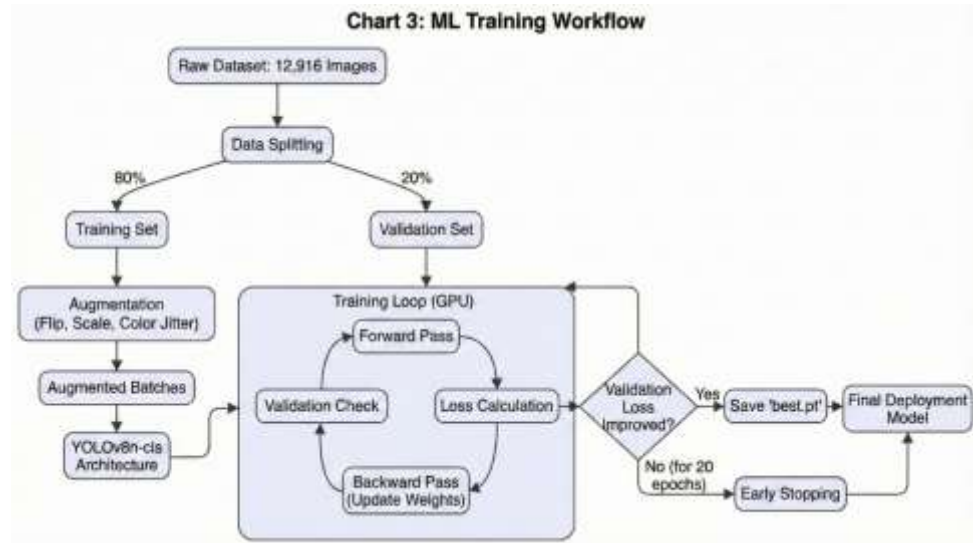


Figure 4: ML Model Training Workflow

4. Experiments and Key Results

4.1 Accuracy Metrics

The model was evaluated on a hold-out validation set of 1,123 images. The results indicate high reliability for industrial sorting.

- **Top-1 Accuracy: 98.7%** (The model's highest probability prediction is correct 98.7% of the time).
- **Top-5 Accuracy: 100%** (The correct label is always within the top 5 predictions—trivial here as we only have 4 classes, but confirms zero "hallucinations").

4.2 Latency Analysis (Per Frame)

Performance profiling was conducted to measure the time cost of each stage in the pipeline.

Table 6: Showing the Latency of detection of the ML Model

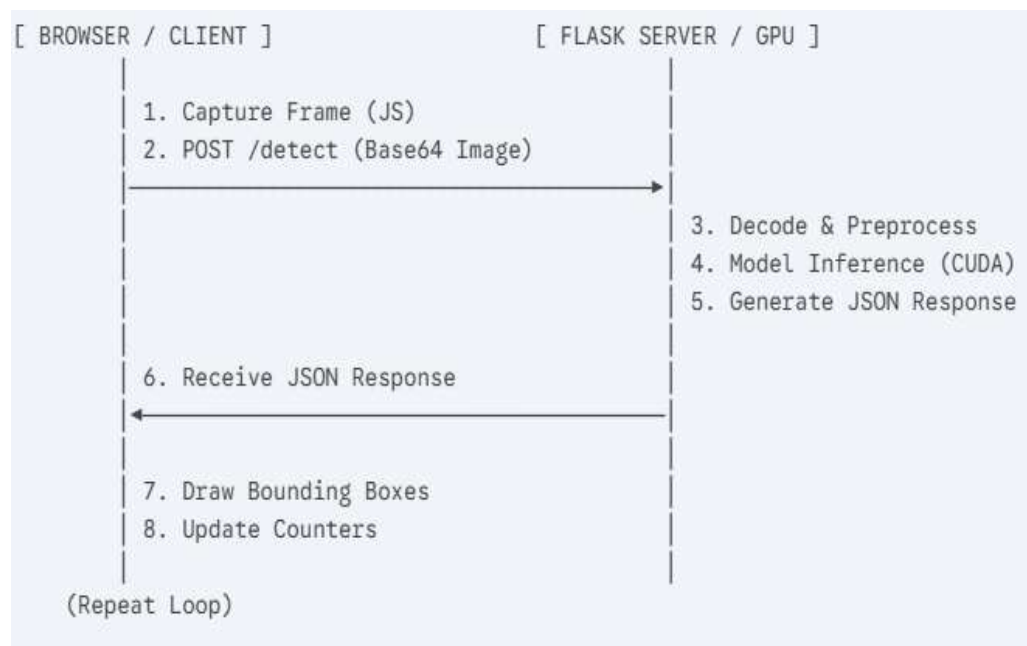
Pipeline Stage	Average Duration	Notes
Preprocessing	~0.1 ms	Base64 decoding and resizing.
Stage 1: Detection	~7.5 ms	Locating all objects in the frame.
Stage 2: Classification	~3.5 ms	Per object classification time.
Post-processing	~0.5 ms	JSON serialization and threshold filtering.
Total Inference Time	~15 - 25 ms	Supports theoretically high FPS.

5. Deployment Environment

The system is deployed via a Client-Server architecture to allow remote accessibility.

- **Server Side:** A Flask (Python) application acts as the API gateway. It receives Base64 encoded images, passes them to the GPU-resident models, and returns JSON data.
- **Client Side:** A Javascript-based frontend captures video from the user's camera (Webcam or DroidCam), renders the video to a canvas, and overlays the detection results.

Network Communication Flow:



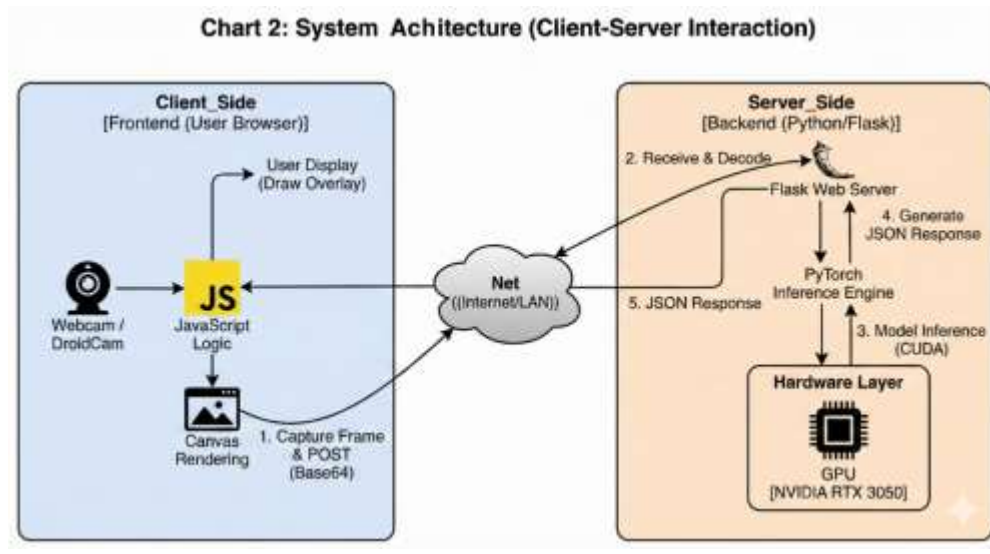


Figure 5: Entire ML System Architecture

6. Limitations and Future Scope

While the system meets the core objectives, specific limitations were identified during the experimental phase:

- **Network Latency:** While local inference is fast (~25ms), network transmission over HTTP can introduce lag, reducing the effective Web FPS to 6–10 FPS depending on Wi-Fi quality.
- **Occlusion Handling:** If two bananas heavily overlap, the Stage 1 detector may merge them into a single bounding box, leading to mixed classification signals.

Future Enhancements:

- **Object Tracking:** Implementation of **ByteTrack** algorithms to assign persistent IDs to fruit, ensuring that a single banana moving across the conveyor belt is counted exactly once.
- **Protocol Optimization:** Migrating from HTTP POST requests to **WebSockets** to establish a persistent full-duplex connection, significantly reducing network overhead and latency.

Sample Codes and Project Structure:

Project Structure:

```

MajorProject/
├── banana_detection/
│   ├── dataset/           # (Optional local copy)
│   ├── models/
│   │   └── banana_classifier.pt  # Trained classification model
│   ├── runs/
│   │   ├── classify/
│   │   │   └── weights/
│   │   │       └── best.pt      # Best model checkpoint
│   └── templates/
│       └── index.html         # Web interface
├── static/
│   └── style.css              # Styling
├── train.py                   # Training script
├── main.py                    # Local webcam detection
├── web_app.py                  # Network-accessible web server
├── setup_project.py           # Project folder setup
├── requirements.txt            # Dependencies
├── ANNOTATION_GUIDE.md        # How to annotate data
└── PROJECT_DOCUMENTATION.md   # Documentation file

```

train.py code:

```

1  """
2  Banana Ripeness Detection - Training Script
3  =====
4  Trains a YOLOv8 classification model on your folder-organized dataset.
5  The trained model will be used to classify detected bananas in live video.
6
7  Dataset Structure:
8  | D:/VSCODE/NEWMAJOR/data/banana/
9  | - train/ (overripe, ripe, rotten, unripe)
10 | - valid/ (same subfolders)
11 | - test/ (same subfolders)
12
13 Usage: python train.py
14 """
15
16 import os
17 import torch
18 from pathlib import Path
19 from ultralytics import YOLO
20
21
22 # =====
23 # CONFIGURATION
24 # =====
25
26 DATASET_PATH = Path(r"D:\VSCODE\NEWMAJOR\data\banana")
27
28 CONFIG = {
29     "model": "yolov8n-cls.pt", # Classification model
30     "epochs": 100,
31     "batch_size": 32,
32     "image_size": 224,
33     "patience": 20,
34     "device": 0,
35     "workers": 4,
36     "project": "banana_detection/runs",
37     "name": "classify",
38 }
39
40
41 def check_gpu():
42     """Check GPU availability."""
43     print("\n" + "="*60)
44     print("GPU CHECK")
45     print("="*60)
46     if torch.cuda.is_available():
47         print(f"✓ GPU: {torch.cuda.get_device_name(0)}")
48         print(f"✓ CUDA: {torch.version.cuda}")
49         return True
50     print("X No GPU - using CPU")
51     return False
52
53
54 def validate_dataset():
55     """Validate dataset structure."""
56     print("\n" + "="*60)
57     print("DATASET")
58     print("="*60)
59
60     classes = ["overripe", "ripe", "rotten", "unripe"]
61     for split in ["train", "valid"]:
62         split_path = DATASET_PATH / split
63         print(f"\n{split.upper()}:")
64         for cls in classes:
65             cls_path = split_path / cls
66             count = len(list(cls_path.glob("*.jpg"))) if cls_path.exists() else 0
67             print(f"    {cls}: {count} images")
68
69     return (DATASET_PATH / "train").exists()
70
71
72 def train():

```

```

71
72 def train():
73     """Train the classification model."""
74     print("\n" + "="*60)
75     print("BANANA RIPENESS CLASSIFIER - TRAINING")
76     print("="*60)
77
78     has_gpu = check_gpu()
79     if not validate_dataset():
80         print("X Dataset not found!")
81         return
82
83     print(f"\n✓ Loading {CONFIG['model']}...")
84     model = YOLO(CONFIG["model"])
85
86     print("\n" + "="*60)
87     print("TRAINING STARTED")
88     print(f"Epochs: {CONFIG['epochs']} | Batch: {CONFIG['batch_size']} | GPU: {has_gpu}")
89     print("="*60 + "\n")
90
91     results = model.train(
92         data=str(DATASET_PATH),
93         epochs=CONFIG["epochs"],
94         batch=CONFIG["batch_size"],
95         imgsz=CONFIG["image_size"],
96         patience=CONFIG["patience"],
97         device=CONFIG["device"] if has_gpu else "cpu",
98         workers=CONFIG["workers"],
99         project=CONFIG["project"],
100        name=CONFIG["name"],
101        exist_ok=True,
102        pretrained=True,
103        verbose=True,
104        plots=True,
105    )
106
107    # Save best model
108
109    # Save best model
110    best = Path(CONFIG["project"]) / CONFIG["name"] / "weights" / "best.pt"
111    if best.exists():
112        import shutil
113        out_dir = Path("banana_detection/models")
114        out_dir.mkdir(exist_ok=True)
115        shutil.copy(best, out_dir / "banana_classifier.pt")
116        print("\n" + "="*60)
117        print("✓ TRAINING COMPLETE!")
118        print(f"✓ Model saved: banana_detection/models/banana_classifier.pt")
119        print("✓ Run 'python main.py' for live detection!")
120        print("="*60)
121
122    return results
123
124 if __name__ == "__main__":
125     train()

```

main.py code:

```
main.py > ...
1 """
2 Banana Ripeness Detection - Live Detection + Classification
3 """
4 Hybrid approach:
5 1. Detect bananas using pre-trained YOLOv8 (COCO dataset includes bananas)
6 2. Classify each detected banana using your trained ripeness model
7
8 This gives you BOUNDING BOXES + RIPPENESS CLASSIFICATION for multiple bananas!
9
10 Controls:
11 q - Quit
12 s - Screenshot
13
14 Usage: python main.py
15 """
16
17 import cv2
18 import time
19 import torch
20 import numpy as np
21 from pathlib import Path
22 from ultralytics import YOLO
23
24
25 # =====
26 # CONFIGURATION
27 # =====
28
29 # Your trained ripeness classifier
30 CLASSIFIER_PATH = Path("banana_detection/models/banana_classifier.pt")
31 CLASSIFIER_FALLBACK = Path("banana_detection/runs/classify/weights/best.pt")
32
33 # Pre-trained detector (COCO has bananas as class 46)
34 DETECTOR_MODEL = "yolov8n.pt" # Will download automatically
35 BANANA_CLASS_ID = 46 # Banana class in COCO dataset
36
37 # Detection settings
38
39 # Detection settings
40 DETECTION_CONF = 0.4
41 CLASSIFICATION_CONF = 0.1
42
43 # GPU
44 DEVICE = 0 if torch.cuda.is_available() else "cpu"
45
46 # Colors for ripeness (BGR)
47 COLORS = {
48     "unripe": (0, 255, 0), # Green
49     "ripe": (0, 255, 255), # Yellow
50     "overripe": (0, 165, 255), # Orange
51     "rotten": (0, 0, 255), # Red
52 }
53
54 WINDOW_NAME = "Banana Ripeness Detection"
55
56 def load_models():
57     """Load detector and classifier models."""
58     print("\n" + "-" * 60)
59     print("LOADING MODELS")
60     print("-" * 60)
61
62     # Load detector (pre-trained on COCO)
63     print(f"Loading banana detector (YOLOv8 COCO)...")
64     detector = YOLO(DETECTOR_MODEL)
65
66     # Load classifier (your trained model)
67     if CLASSIFIER_PATH.exists():
68         classifier_path = CLASSIFIER_PATH
69     elif CLASSIFIER_FALLBACK.exists():
70         classifier_path = CLASSIFIER_FALLBACK
71     else:
72         print(f"Classifier not found! Run 'python train.py' first.")
73         print(f"Expected: {CLASSIFIER_PATH}")
74         return None, None
```

```

72     print(f" Expected: {CLASSIFIER_PATH}")
73     return None, None
74
75 print(f"/ Loading ripeness classifier: {classifier_path}")
76 classifier = VOLO(str(classifier_path))
77
78 if torch.cuda.is_available():
79     print(f"/ GPU: {torch.cuda.get_device_name(0)}")
80
81 return detector, classifier
82
83
84 def classify_banana(classifier, banana_img):
85     """Classify a cropped banana image."""
86     results = classifier.predict(source=banana_img, device=DEVICE, verbose=False)
87
88     if results[0].probs is not None:
89         probs = results[0].probs
90         class_id = probs.top1
91         confidence = probs.top1conf.item()
92         class_name = results[0].names[class_id]
93         return class_name, confidence
94
95     return "unknown", 0.0
96
97
98 def process_frame(frame, detector, classifier):
99     """Detect bananas and classify each one."""
100     # Detect objects
101     results = detector.predict(
102         source=frame,
103         conf=DETECTION_CONF,
104         device=DEVICE,
105         verbose=False,
106         classes=BANANA_CLASS_ID) # Only detect bananas
107
108

```

```

109     CLASS_ID = BANANA_CLASS_ID) # Only detect bananas
110 }
111
112 detections = []
113
114 if results[0].boxes is not None:
115     boxes = results[0].boxes
116
117     for box in boxes:
118         # Get bounding box
119         x1, y1, x2, y2 = box.boxes[0].cpu().numpy().astype(int)
120         det_conf = float(box.conf[0].cpu().numpy())
121
122         # Crop banana region
123         x1, y1 = max(0, x1), max(0, y1)
124         x2, y2 = min(frame.shape[1], x2), min(frame.shape[0], y2)
125
126         if x2 > x1 and y2 > y1:
127             banana_crop = frame[y1:y2, x1:x2]
128
129             # Classify ripeness
130             ripeness, cls_conf = classify_banana(classifier, banana_crop)
131
132             detections.append(
133                 {
134                     "box": (x1, y1, x2, y2),
135                     "ripeness": ripeness,
136                     "det_conf": det_conf,
137                     "cls_conf": cls_conf,
138                 })
139
140     return detections
141
142
143 def draw_detections(frame, detections):
144     """Draw bounding boxes and labels."""
145     for det in detections:
146         x1, y1, x2, y2 = det["box"]

```

```

141     for det in detections:
142         x1, y1, x2, y2 = det["box"]
143         ripeness = det["ripeness"]
144         cls_conf = det["cls_conf"]
145
146         # Get color
147         color = COLORS.get(ripeness.lower(), (255, 255, 255))
148
149         # Draw box
150         cv2.rectangle(frame, (x1, y1), (x2, y2), color, 3)
151
152         # Draw label
153         label = f"{ripeness}: {cls_conf:.0%}"
154         label_size = cv2.getTextSize(label, cv2.FONT_HERSHEY_SIMPLEX, 0.7, 2)[0]
155
156         # label background
157         cv2.rectangle(frame, (x1, y1 - 30), (x1 + label_size[0] + 10, y1), color, -1)
158         cv2.putText(frame, label, (x1 + 5, y1 - 8), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0, 0), 2)
159
160     return frame
161
162
163 def draw_overlay(frame, fps, detection_count):
164     """Draw FPS and detection info."""
165     h, w = frame.shape[:2]
166
167     # Top info bar
168     overlay = frame.copy()
169     cv2.rectangle(overlay, (0, 0), (200, 30), (0, 0, 0), -1)
170     cv2.addWeighted(overlay, 0.0, frame, 0.4, 0, frame)
171
172     cv2.putText(frame, f"FPS: {fps:.0f}", (15, 25), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0), 2)
173     cv2.putText(frame, f"Bananas: {detection_count}", (15, 50), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0), 2)
174
175     # Legend
176     legend_y = h - 100

```

```

209     print("X Cannot open webcam!")
210     return
211
212 cap.set(cv2.CAP_PROP_FRAME_WIDTH, 1280)
213 cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 720)
214
215 print("✓ Starting detection...")
216 print("\n" + "*" * 60)
217 print("Press 'q' to quit | Press 's' for screenshot")
218 print("*" * 60 + "\n")
219
220 # FPS tracking
221 fps = 0
222 frame_count = 0
223 start_time = time.time()
224 screenshot_num = 0
225
226 while True:
227     ret, frame = cap.read()
228     if not ret:
229         break
230
231     # Detect and classify
232     detections = process_frame(frame, detector, classifier)
233
234     # Draw results
235     frame = draw_detections(frame, detections)
236
237     # FPS calculation
238     frame_count += 1
239     if time.time() - start_time > 0:
240         fps = frame_count / (time.time() - start_time)
241     if frame_count > 30:
242         frame_count = 0
243         start_time = time.time()
244
245     return

```

```

243:         start_time = time.time()
244:
245:         # Draw overlay
246:         frame = draw_overlay(frame, fps, len(detections))
247:
248:         # Show
249:         cv2.imshow(WINDOW_NAME, frame)
250:
251:         # Key handling
252:         key = cv2.waitKey(1) & 0xFF
253:         if key == ord('q'):
254:             break
255:         elif key == ord('s'):
256:             cv2.imwrite(f'screenshot_{screenshot_num:03d}.jpg', frame)
257:             print(f'✓ Saved screenshot_{screenshot_num:03d}.jpg')
258:             screenshot_num += 1
259:
260:     cap.release()
261:     cv2.destroyAllWindows()
262:     print("\n✓ Done!")
263:
264:
265: if __name__ == "__main__":
266:     main()
267:

```

web_app.py code:

```

web_app.py > ...
1  """
2  Banana Ripeness Detection - Web Application
3  =====
4  Flask-based web server for real-time banana detection using phone camera.
5
6  Usage:
7  | python web_app.py
8
9  Then open http://your-ip:3000 on your phone browser.
10 """
11
12 import base64
13 import io
14 import socket
15 import numpy as np
16 import socket
17 from pathlib import Path
18 from flask import Flask, render_template, request, jsonify
19 from flask_cors import CORS
20 from PIL import Image
21 from ultralytics import YOLO
22
23 # =====
24 # Configuration
25 # =====
26
27 CLASSIFIER_PATH = Path("banana_detection/models/banana_classifier.pt")
28 CLASSIFIER_FALLBACK = Path("banana_detection/runs/classify/weights/best.pt")
29 DETECTOR_MODEL = "yolo10n.pt"
30 BANANA_CLASS_ID = 45 # Banana in COCO
31 FRUIT_CLASSES = [46, 47, 49, 50, 51, 52] # banana, apple, orange, broccoli, carrot, hot dog (for testing)
32
33 DETECTION_CONF = 0.20 # Very low for better detection
34 DEVICE = 0 if torch.cuda.is_available() else "cpu"
35 DEBUG = True
36
37 # Flask app
38 app = Flask(__name__)
39 CORS(app)
40
41 # Global models (loaded once)
42 detector = None
43 classifier = None
44
45 classifier = None
46
47 def load_models():
48     """Load detection and classification models."""
49     global detector, classifier
50
51     print("Loading models...")
52
53     # Detector
54     detector = YOLO(DETECTOR_MODEL)
55     print(f"✓ Detector loaded: {DETECTOR_MODEL}")
56
57     # Classifier
58     if CLASSIFIER_PATH.exists():
59         classifier = YOLO(str(CLASSIFIER_PATH))
60     elif CLASSIFIER_FALLBACK.exists():
61         classifier = YOLO(str(CLASSIFIER_FALLBACK))
62     else:
63         raise FileNotFoundError("Classifier model not found! Run train.py first.")
64
65     print(f"✓ Classifier loaded")
66
67     if torch.cuda.is_available():
68         print(f"✓ GPU: {torch.cuda.get_device_name(0)}")
69
70 def get_local_ip():
71     """Get the local IP address for LAN access."""
72     try:
73         s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
74         s.connect(("8.8.8.8", 80))
75         ip = s.getsockname()[0]
76         s.close()
77         return ip
78     except:
79         return "127.0.0.1"
80
81 def process_image(image_data):
82     """Process base64 image - detect objects and classify as bananas"""
83     # Decode base64 image
84     image_bytes = base64.b64decode(image_data.split(",")[1])

```

```

84 # Decode base64 image
85 image_bytes = base64.b64decode(image_data.split(",")[1])
86 image = Image.open(io.BytesIO(image_bytes)).convert("RGB")
87 img_array = np.array(image)
88 h, w = img_array.shape[:2]
89
90
91
92 # Detect ALL objects (not just bananas) with low confidence
93 results = detector.predict(
94     source=img_array,
95     conf=DETECTION_CONF,
96     device=DEVICE,
97     verbose=False
98     # No class filter - detect everything
99 )
100
101 if results[0].boxes is not None and len(results[0].boxes) > 0:
102     if DEBUG:
103         detected_names = [detector.names[int(c)] for c in results[0].boxes.cls.cpu().numpy()]
104         print(f"Found: {detected_names}")
105
106     for box in results[0].boxes:
107         x1, y1, x2, y2 = box.xyxy[0].cpu().numpy().astype(int).tolist()
108         x1c, y1c = max(0, x1), max(0, y1)
109         x2c, y2c = min(w, x2), min(h, y2)
110
111         if x2c > x1c and y2c > y1c:
112             crop = img_array[y1c:y2c, x1c:x2c]
113
114             # Classify each detected object
115             cls_results = classifier.predict(source=crop, device=DEVICE, verbose=False)
116
117             if cls_results[0].probs is not None:
118                 probs = cls_results[0].probs
119                 class_name = cls_results[0].names[probs.top1]
120                 confidence = float(probs.top1conf)
121
122                 # Only keep if classifier is confident it's a banana type
123                 if confidence > 0.5:
124                     detections.append([
125                         "box": [x1, y1, x2, y2],
126
127                         "box": [x1, y1, x2, y2],
128                         "class": class_name,
129                         "confidence": confidence
130                     ])
131
132 # Fallback: If no objects detected, classify entire frame
133 # But first check if frame has actual content (not just black/dark)
134 if len(detections) == 0:
135     # Check average brightness - skip if too dark
136     avg_brightness = np.mean(img_array)
137     if avg_brightness > 30: # Only classify if image has content
138         cls_results = classifier.predict(source=img_array, device=DEVICE, verbose=False)
139
140         if cls_results[0].probs is not None:
141             probs = cls_results[0].probs
142             class_name = cls_results[0].names[probs.top1]
143             confidence = float(probs.top1conf)
144
145             # Require higher confidence for full-frame fallback
146             if confidence > 0.6:
147                 margin = 30
148                 detections.append([
149                     "box": [margin, margin, w - margin, h - margin],
150                     "class": class_name,
151                     "confidence": confidence
152                 ])
153
154 return detections
155
156
157 # =====
158 # Routes
159 # =====
160
161 @app.route("/")
162 def index():
163     """Serve the main page."""
164     return render_template("index.html")
165
166 @app.route("/detect", methods=["POST"])
167 def detect():

```

```

154
155 @app.route("/detect", methods=['POST'])
156 def detect():
157     """Detection API endpoint."""
158     try:
159         data = request.json
160         image_data = data.get("image")
161
162         if not image_data:
163             return jsonify({"error": "No image provided"}), 400
164
165         detections = process_image(image_data)
166         return jsonify({"detections": detections})
167
168     except Exception as e:
169         return jsonify({"error": str(e)}), 500
170
171
172 # =====
173 # Main
174 # =====
175
176 if __name__ == "__main__":
177     load_model()
178
179     ip = get_local_ip()
180     port = 5000
181
182     print("\n" + "-" * 50)
183     print("BANANA DETECTION WEB SERVER (HTTPS)")
184     print("-" * 50)
185     print(f"Local: https://localhost:{port}")
186     print(f"Network: https://{ip}:{port}")
187     print("Open the Network URL on your phone!")
188     print("Note: Accept the security warning (self-signed certificate)")
189     print("-" * 50 + "\n")
190
191     # Run with HTTPS (required for camera access on phones)
192     app.run(host="0.0.0.0", port=port, debug=False, threaded=True, ssl_context="adhoc")
193
194

```

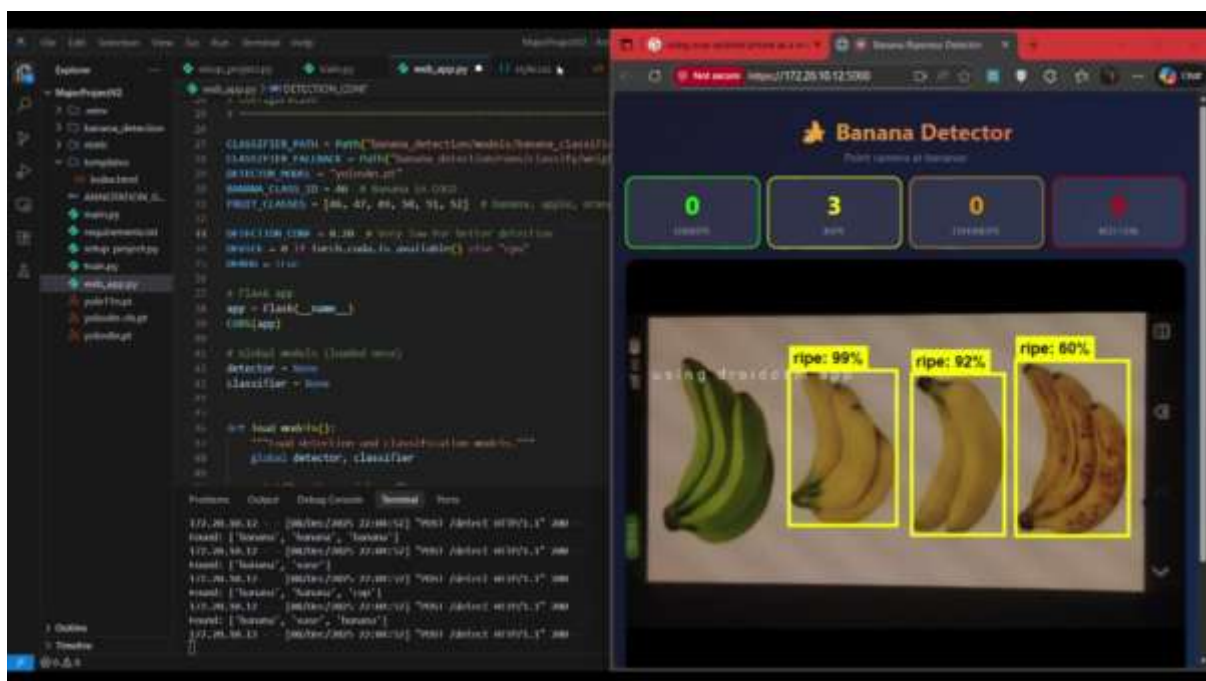


Figure 6: Screenshot of working ML Model.

Chapter 6: Preliminary Arduino Architecture for Mechanical Arm

6.1 Overview of the Control Loop

The physical sorting mechanism is governed by a **Master-Slave architecture**. The Computer Vision system (Master) running on the PC performs high-level decision-making, while the Arduino Microcontroller (Slave) executes low-level physical actuation. This separation ensures that the heavy computational load of the neural network does not interfere with the precise timing required for motor control.

6.2 Serial Communication Protocol (Python to Arduino)

Communication is established via a **UART (Universal Asynchronous Receiver-Transmitter)** serial interface over USB. To minimize latency, we utilize a lightweight single-byte protocol.

- **Library:** Python pyserial library.
- **Baud Rate:** 9600 bps (Bits Per Second).
- **Data Integrity:** A handshake delay of 2 seconds is implemented upon initialization to allow the Arduino bootloader to settle.

Table 7: Showing various serial payloads attached to different fruit class

Detected Class	Serial Payload (Char)	ASCII Value	Meaning
Rotten	'R'	82	REJECT: Defect detected. Trigger actuation.
Unripe	'G'	71	PASS: Good fruit. Remain Idle.
Ripe	'G'	71	PASS: Good fruit. Remain Idle.
Overripe	'G'	71	PASS: Good fruit. Remain Idle.

6.3 Electronic Circuitry and Power Distribution

The actuation system requires significantly higher current than the logic system. Therefore, the power domains are isolated to prevent "Brown-out" resets on the Arduino.

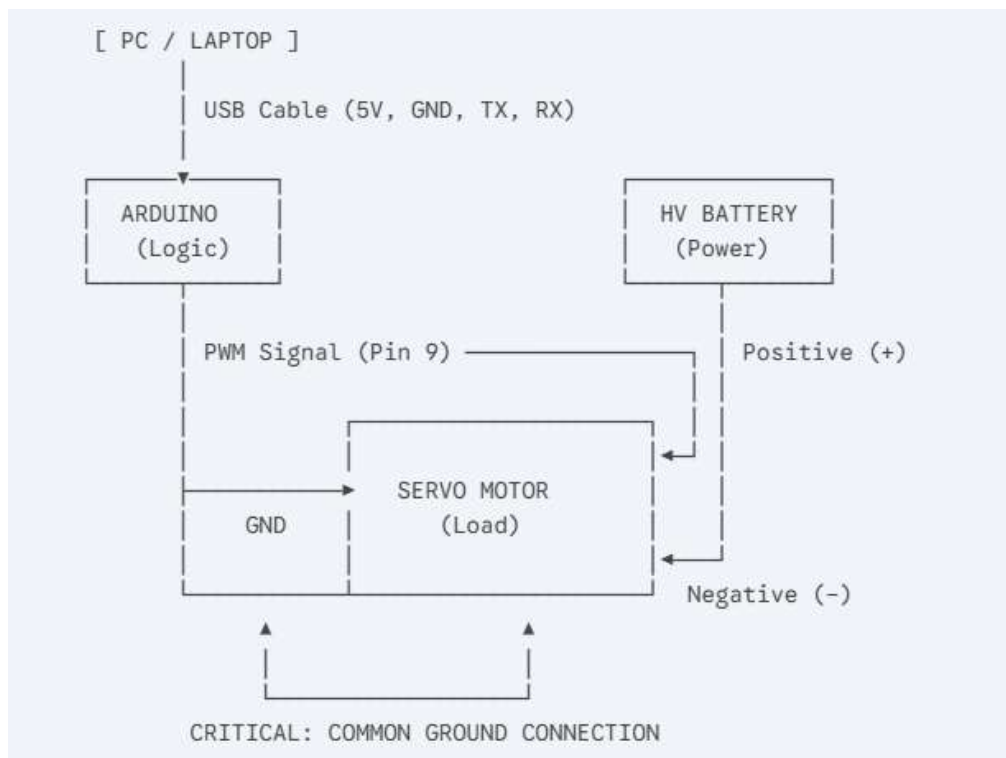
6.3.1 Component Specifications

- **Microcontroller:** Arduino UNO R3 (ATmega328P).
- **Actuator:** MG995 Metal Gear High-Torque Servo Motor.

- *Torque*: 10 kg.cm (Required to push heavy fruit).
- *Operating Voltage*: 4.8V – 7.2V.
- **Power Source**: High-Voltage (HV) External Battery Pack (e.g., 2S Li-Po 7.4V or 9V Battery with Buck Converter).
- **Mechanical Arm**: 3D printed linear arm attached to the servo horn.

6.3.2 Wiring Topology (Common Ground)

A critical requirement for this circuit is the **Common Ground (GND)** connection. While the Positive (+) lines are separated to handle different voltages, the Ground lines must be connected to complete the logic signal loop.



6.4 Microcontroller Logic and Actuation

The Arduino runs a continuous loop monitoring the serial buffer. The servo logic is designed to be **non-blocking** where possible, though a simple delay is used here for mechanical stability during the "sweep" action.

The Sorting Algorithm:

1. **Idle State**: Servo maintains a 0 degree angle (parallel to the belt), allowing fruit to pass.
2. **Trigger State**: Upon receiving byte 'R':
 - Servo rotates rapidly to 90 degree (blocking the path/pushing fruit).
 - System holds position for **2 seconds** (configurable dwell time).
 - Servo returns to 0 degree.
3. **Buffer Clearing**: The serial buffer is flushed after actuation to prevent a queue of old commands from executing late.

6.5 System Flowcharts

6.5.1 Data & Signal Flow

This chart illustrates the transformation of a visual detection into a physical movement.

Arduino and Servo Motor Communication

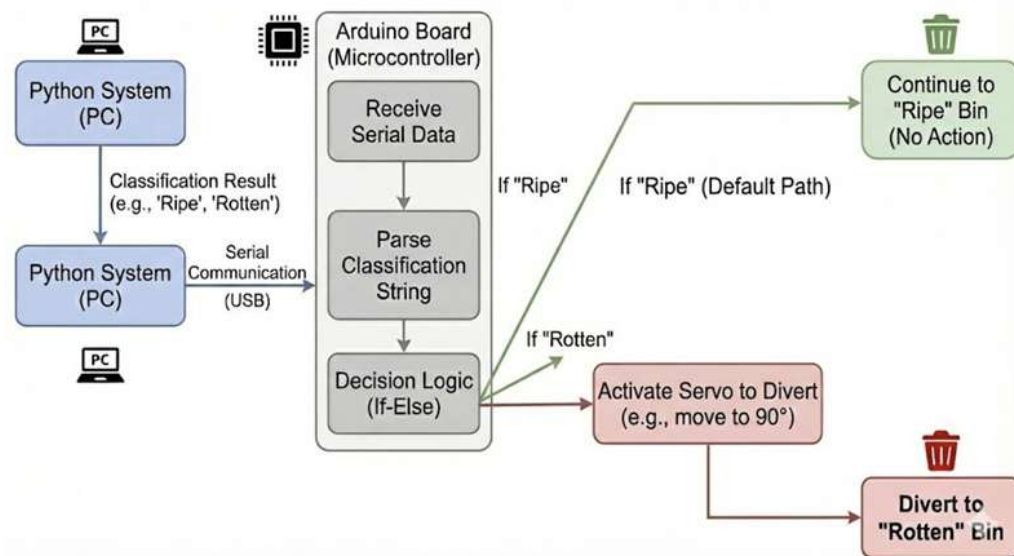


Figure 7: Arduino Servo Motor Communication

6.6 Deployment Considerations

- **Voltage Regulation:** Since the HV battery might exceed the servo's 7.2V limit (e.g., a 9V battery), a **Buck Converter (LM2596)** should be placed between the battery and the servo to regulate voltage to a safe 6V, ensuring consistent torque without overheating.
- **Electrical Noise:** Servos create electrical noise. A 470 microF electrolytic capacitor is placed across the servo power rails to filter noise and prevent the Arduino from resetting due to voltage spikes.
- **Safety:** The mechanical arm moves with significant force. The code implements a "Soft Start" in the setup() phase to prevent the arm from jerking violently when power is first applied.

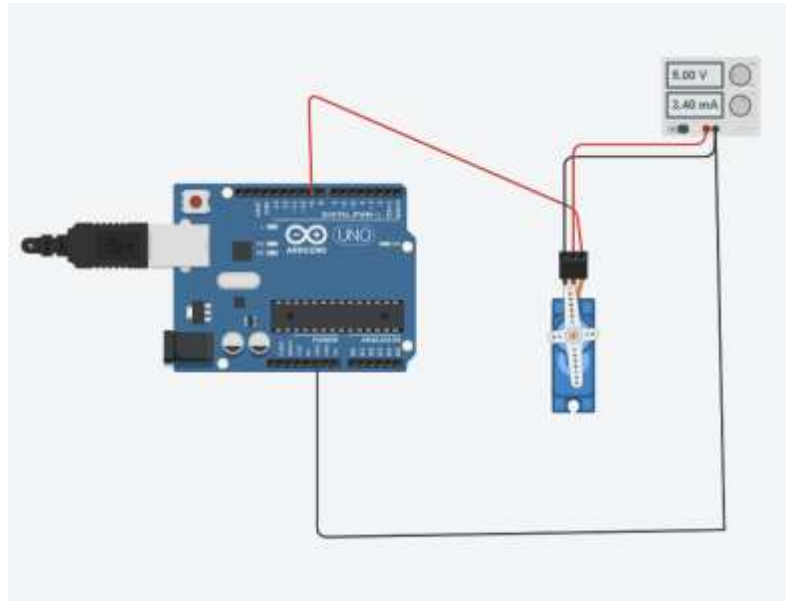


Figure 8: Arduino and Servo Motor Connections.

Arduino Code:

```

1  #include <Servo.h>
2
3  Servo servoMotor;
4  String inputData = "";
5
6  void setup() {
7      Serial.begin(9600);
8      servoMotor.attach(9);
9      servoMotor.write(0);
10     Serial.println("Enter true or false:");
11 }
12
13 void loop() {
14     if (Serial.available() > 0) {
15         inputData = Serial.readStringUntil('\n');
16         inputData.trim();
17
18         if (inputData == "true") {
19             servoMotor.write(90);
20             Serial.println("Spoilt fruit detected → Servo at 90°");
21         }
22         else if (inputData == "false") {
23             servoMotor.write(0);
24             Serial.println("Good fruit detected → Servo at 0°");
25         }
26         else {
27             Serial.println("Invalid input. Type true or false.");
28         }
29
30         delay(1000);
31     }
32 }
33

```

Chapter 7: Conclusion of Major Project Phase 1 and Future Scopes

7.1 Summary of Work Completed

Significant progress has been made in bridging the gap between theoretical design and a functional, intelligent prototyping system. We have successfully moved beyond basic computer vision techniques to implement a state-of-the-art Deep Learning pipeline.

Key milestones achieved to date include:

- **Advanced ML Pipeline Implementation:** We have successfully developed and deployed a **Two-Stage Cascade Architecture** using YOLOv8. This includes a detection layer (YOLOv8n) for localization and a custom-trained classification layer (YOLOv8n-cls).
- **Model Training and Validation:** The classifier has been trained on a dataset of **12,916 images**, achieving a **Top-1 Accuracy of 98.7%** in distinguishing between Unripe, Ripe, Overripe, and Rotten bananas.
- **System Integration:** A robust **Client-Server architecture** has been established using Flask and JavaScript, enabling real-time inference at 25+ FPS locally and 6–10 FPS over a network interface.
- **Hardware-Software Interfacing:** The communication bridge between the Python backend and the embedded hardware has been finalized. We have implemented a stable **UART Serial Protocol** that successfully transmits classification payloads (e.g., 'R' for Rotten) to the microcontroller.
- **Actuation Logic:** The Arduino firmware has been written and validated to control the high-torque **MG995 servo motor**, including the implementation of non-blocking logic and power isolation circuits to handle high-current loads safely.

7.2 Future Work and Next Steps

With the software logic, machine learning models, and electronic control circuits fully operational, the final phase of the project will focus on physical assembly, field testing, and optimization for industrial scalability.

The next steps may include:

1. **Physical Assembly & Calibration:** Complete the 3D printing of the conveyor frame and mechanical arm. The servo motor and Arduino will be mounted to the chassis, ensuring the mechanical arm is calibrated to the precise angle (90°) required to divert fruit without causing damage.
2. **Environmental Stress Testing:** Conduct full-system trials under varying lighting conditions and background noise to ensure the YOLOv8 detector remains robust outside of controlled laboratory environments.
3. **Implementation of Object Tracking:** To improve counting accuracy, we plan to integrate the **ByteTrack** algorithm. This will assign unique IDs to each fruit, preventing double-counting as the fruit moves across the camera's field of view.

4. **Edge Deployment:** While the current system runs on a laptop GPU (RTX 3050), future work involves optimizing the models (via TensorRT) for deployment on embedded edge devices like the **NVIDIA Jetson Nano** or **Raspberry Pi 5**, creating a standalone, compact sorting unit.

Major Project Phase - 2

Chapter 8: Multi-Fruit Expansion and ML Optimization

8.1 Broadening the Classification Scope

While Phase 1 focused exclusively on banana ripeness, Phase 2 extends the system's utility to a multi-product environment. We have integrated Apples and Oranges into the inference pipeline. This requires a transition from a binary "Good/Rotten" logic to a multi-class hierarchical model.

- Apples: Focused on detecting surface bruises and "bitter pit" using localized texture variances.
- Oranges: Identification of "green mold" and rind breakdown, which often present subtle color shifts but distinct texture changes.

8.2 Texture-Based Feature Extraction

To improve classification accuracy beyond simple color thresholding, we have implemented Texture-Specific Training. Traditional CNNs can sometimes be "fooled" by lighting; however, textures like the roughness of a rotten orange or the waxy smoothness of a fresh apple are deterministic.

Technologies Utilized:

- GLCM (Gray-Level Co-occurrence Matrix): To calculate spatial relationship of pixels, extracting features like *Contrast*, *Correlation*, and *Homogeneity*.
- LBP (Local Binary Patterns): Used for robust local texture description. The LBP operator is defined as:

$$LBP_{P,R} = \sum_{p=0}^{P-1} s(g_p - g_c) 2^p$$

where g_c is the gray value of the center pixel and g_p are the values of P surrounding pixels at radius R .

Chapter 9: Advanced Electronic Sub-System

9.1 Transition to Arduino Mega 2560

In Phase 1, an Arduino Uno was utilized. For Phase 2, we have upgraded to the **Arduino Mega 2560**. This transition provides:

- **Increased SRAM:** 8KB vs 2KB, allowing for larger serial buffers to prevent data loss during high-speed sorting.
- **Expanded PWM Channels:** Crucial for future expansion (e.g., adding multiple sorting arms or sensors).

9.2 Power Management and External 5V 3A Supply

The **MG995 High-Torque Servo** can draw up to **1.2A** under stall conditions. Powering this directly from the Arduino's 5V pin would cause "Brown-outs" or permanent damage to the board's voltage regulator.

Assembly Configuration:

1. **External Supply:** A 5V 3A DC adapter powers the servo via a breadboard power rail.
2. **Logic Link:** The Arduino Mega is powered via USB or a separate 7-12V DC jack.
3. **Common Ground:** The negative terminal of the 5V supply **must** be connected to the Arduino GND pin to provide a reference for the PWM signal.

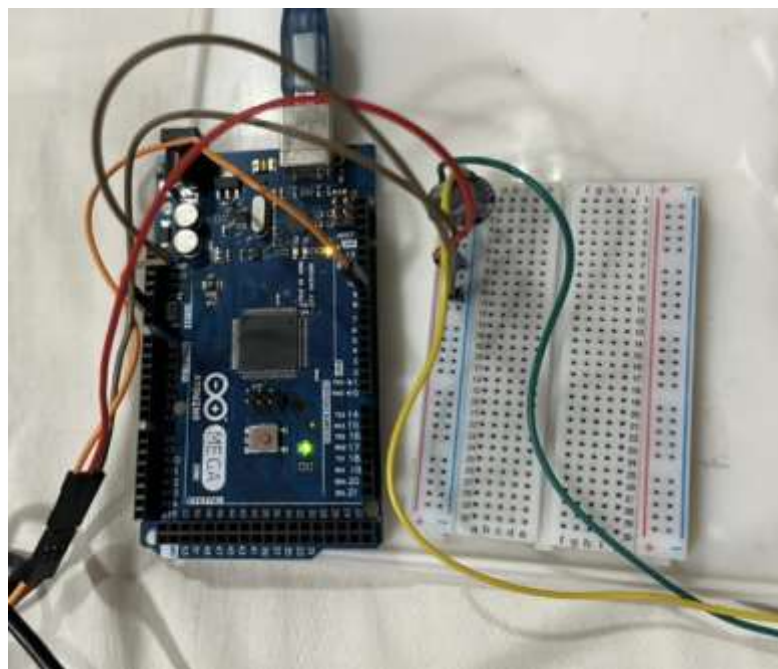


Figure 9: Arduino Mega and breadboard connection with servo and power supply.

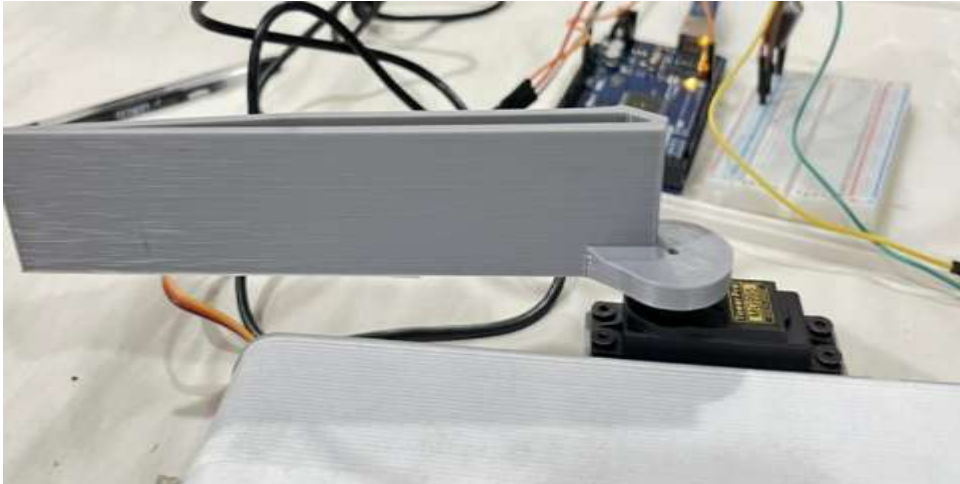


Figure 10: Servo MG995 with lever mounted on it connected to Arduino.



Figure 11: 5V 3A DC supply for Servo Motor and 12V 2A DC Supply for DC Motor

Chapter 10: Mechanical Design and Physical Architecture

10.1 Introduction and Design Philosophy

The mechanical subsystem serves as the physical backbone of the **Automated Quality Control System**. While the Machine Learning model provides the "intelligence," the mechanical assembly must provide the "execution"—ensuring that fruits are transported smoothly and diverted accurately without incurring mechanical damage or bruising.

The design follows a **Modular Ladder-Frame Architecture**, optimized for a small-scale agricultural sorting environment. By utilizing a high-strength-to-weight ratio design, the system minimizes the power requirements of the drive motors while maintaining the rigidity necessary for high-speed automated sorting.

10.2 Integrated Design and Fabrication Workflow

To transition from theoretical concepts to the functional assembly seen in the physical prototype, a specialized digital-to-physical workflow was adopted:

- **CAD Environment:** All components were modeled using **Fusion 360**, utilizing **Synchronous Modeling** and **Parametric Design** to ensure perfect alignment between the drive rollers, idler rollers, and the chassis.
- **Material Selection:** The primary structural members (grey rails and cross-supports) were fabricated using **Polylactic Acid (PLA)** via FDM 3D printing, chosen for its dimensional stability and ease of rapid prototyping.
- **Surface Engineering:** The drive roller (white) features a **studded surface texture**, a critical design choice intended to increase the coefficient of friction between the roller and the conveyor belt, preventing slippage during the high-torque actuation of the DC motors.

10.3 Functional Assembly Overview

As shown in the assembled prototype, the system consists of three functional zones:

1. **The Drive Zone:** Powered by dual **12V DC Gear Motors**, this zone provides the primary motive force to the textured drive roller.
2. **The Transport Zone:** A series of three precision-aligned idler rollers maintain belt tension and provide a flat surface for the fruits (bananas, apples, or oranges) to be captured by the camera for YOLOv8 inference.
3. **The Actuation Mounting Zone:** Integrated brackets on the chassis are positioned to house the **MG995 servo-operated arm**, ensuring the "Decision Execution Unit" has a stable pivot point for fruit diversion.



Figure 12: Assembled 3D printed outer skeleton with 2 12V DC Motors attached.

10.4 Main Structural Assembly: Chassis and Base Infrastructure

The structural integrity of the sorting system relies on a three-part interlocking chassis. This assembly consists of two longitudinal side rails and four transverse base plates, creating a rigid "Ladder Frame" architecture.

10.4.1 Primary Side Rails (Part 1 & Part 2)

The side rails (Size: 300 mm* 47.82 mm) serve as the primary mounting interface for all active and passive components.

- **Design and Purpose:**
 - **Part 1 (Passive Rail):** Acts as the supporting wall for the idler rollers and drive motor. It features precision-drilled holes for shaft alignment.
 - **Part 2 (Active/Servo Rail):** This rail includes an integrated **Servo Mounting Provision**. The vertical tabs are engineered specifically to secure the **MG995 High-Torque Servo**. This ensures that the reactive force generated during the "sweeping" motion is absorbed by the chassis frame rather than the motor's output shaft.
 - **Notch System:** Both rails feature rectangular cutouts along the bottom edge designed to receive the tabs of the base plates, ensuring a self-aligning assembly process.

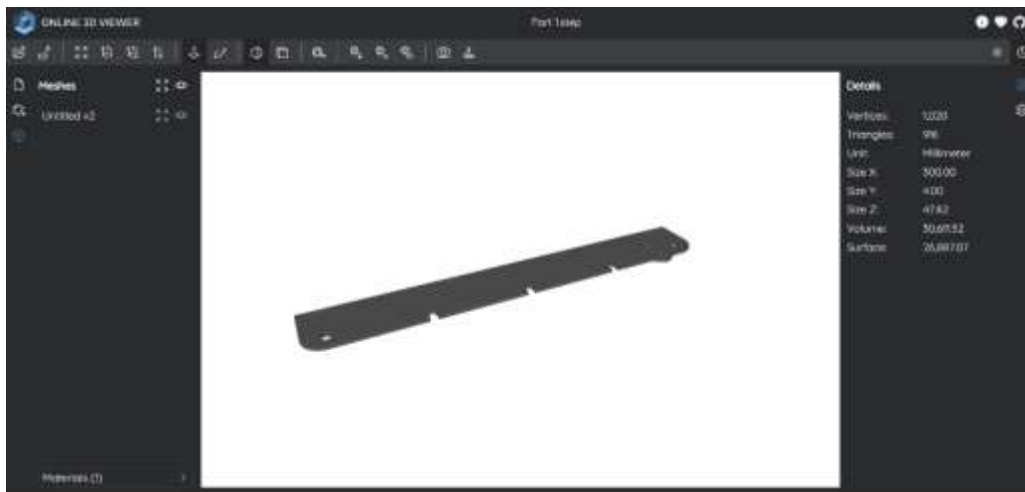


Figure 13: Side Rail 1



Figure 14: Side Rail 2 having provision for servo fixation

10.4.2 Transverse Base Plates (Plate 4qty.step)

These plates (Size: 164 mm * 30 mm) are the critical cross-members that establish the fixed width of the conveyor system.

- **Design and Purpose:**
 - **Width Calibration:** The 164 mm length is optimized to accommodate the width of the rubber conveyor belt while providing enough lateral clearance for fruit transport.
 - **Tab-and-Slot Joint:** The design utilizes a **mechanical interlocking tab** at both ends. When inserted into the side rails, these tabs create a multi-planar joint that significantly increases the assembly's resistance to torsion and "wracking" caused by the 12V DC gear motors.
 - **Low Center of Gravity:** By placing these plates at the bottom of the rails, the overall center of mass is lowered, increasing the system's stability during rapid sorting actuations.

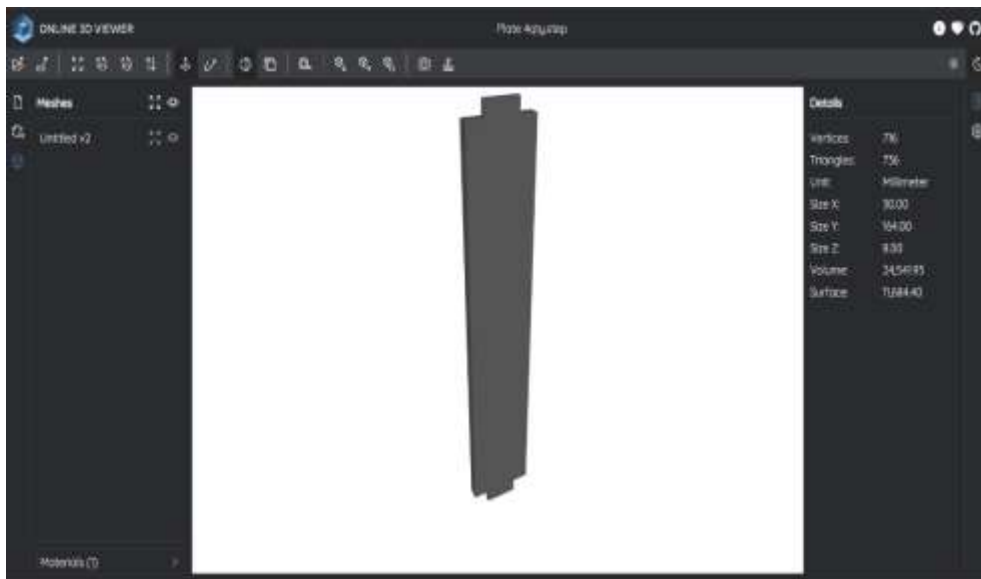


Figure 15: Transverse Base Plates

10.4.3 Autodesk Fusion 360 Feature Breakdown

The design was executed using a parametric workflow to ensure that any change in belt width would automatically update all components.

Feature/Command	Application	Mechanical Benefit
Sketch Constraints	Defining the 300 mm length and tab positions.	Ensures that the notches on the rails perfectly align with the tabs on the plates.
Extrude (Join/New Body)	Creating the 4 mm rails and 8 mm thick base plates.	Optimized thickness for FDM printing to prevent structural warping.
Project (P)	Transferring hole locations from Part 1 to Part 2.	Guarantees that the rollers are perfectly perpendicular to the belt path.
Fillet (F)	Rounded corners on the rails (Size: approx 5 mm).	Eliminates sharp edges for safer handling and reduces stress risers at joint intersections.
Combine (Cut)	Used to subtract the plate tab geometry from the side rails.	Created the perfect "friction fit" interlocking notches seen in the screenshots.

Table 8: Showing various commands used to create the above design

10.5 Conveyor Roller Dynamics: Main and Auxiliary Units

The conveyor system utilizes a combination of active and passive rollers to achieve synchronized fruit transport. These components are critical for maintaining a consistent belt speed, which directly impacts the accuracy of the YOLOv8 detection timing.

10.5.1 Main Studded Drive Roller (H1 Roller 1qty.step)

The H1 Roller is the primary power transmission element of the mechanical assembly. Located at the head of the conveyor, it is directly coupled to the dual 12V DC gear motors.

- **Design Intent and Purpose:** Unlike standard smooth rollers, this unit features a specialized studded surface geometry. These small protrusions are engineered to increase the coefficient of friction and provide a mechanical interlock with the rubber conveyor belt. This prevents axial and longitudinal slippage, ensuring that the motor torque is efficiently converted into linear belt motion even under the weight of multiple fruit payloads.
- **Technical Dimensions:**
 - **Main Diameter (Size X):** 22.44 mm
 - **Total Length (Size Y):** 84.00 mm
 - **Effective Height (Size Z):** 23.60 mm
- **Fusion 360 Design Workflow:**
 - **Revolve:** The core cylindrical body was created by revolving a precision-sketched rectangular profile 360 degrees around the central axis.
 - **Construct Tangent Plane:** A plane was established on the surface of the cylinder to sketch the individual stud profiles.
 - **Extrude (Join):** The studs were extruded from the tangent plane into the main body to create a monolithic structure.
 - **Circular Pattern:** This command was used to replicate the stud feature evenly around the circumference, ensuring the roller remains dynamically balanced during high-speed rotation.

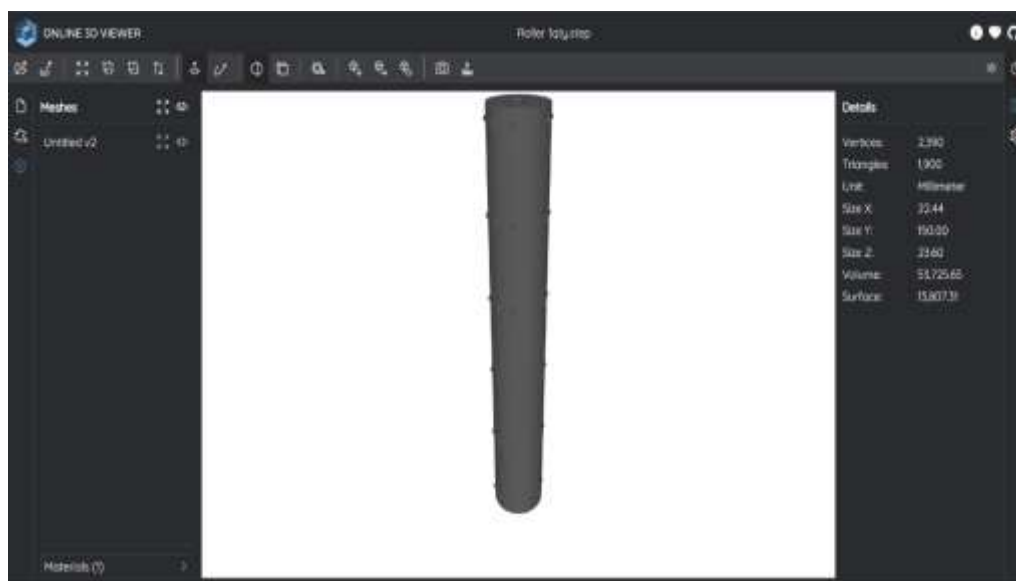


Figure 16: Main Rollers responsible for conveyor belt rotation by providing friction

10.5.2 Auxiliary Idler Rollers (Side roller 6 qty.step)

The transport zone features a series of smooth auxiliary rollers that act as the structural support for the belt's upper run.

- **Design Intent and Purpose:** These rollers are passive and designed for low rotational inertia to minimize the load on the drive system. Their primary purpose is to maintain a perfectly horizontal belt surface. This is essential for the Computer Vision system, as it provides a consistent background and focal plane for the fruits as they move toward the camera. The smooth finish minimizes friction on the underside of the belt, reducing energy consumption and mechanical wear.
- **Technical Dimensions:**
 - **Diameter (Size X/Z):** 11.98 mm
 - **Length (Size Y):** 82.00 mm
- **Fusion 360 Design Workflow:**
 - **Extrude:** The main body was created by extruding an 11.98 mm circular sketch to a length of 82.00 mm.
 - **Fillet:** High-precision fillets were applied to the edges of the roller to ensure smooth belt transitions and prevent edge-wear or snagging.
 - **Hole Command:** A central bore was modeled to accommodate the steel axle pins that secure the rollers to the side rails, ensuring concentric rotation.

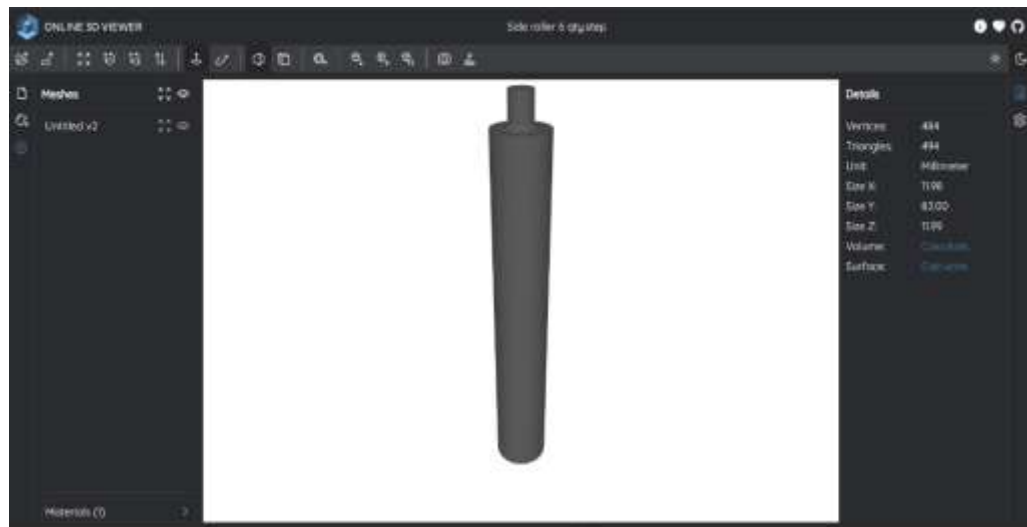


Figure 17: Auxiliary Rollers to prevent slacking of belt while carrying fruits.

10.5.3 Assembly Integration Logic

The rollers are strategically spaced along the 300 mm side rails to prevent belt "sag" when a heavy payload passes through the inspection zone. The main drive roller is deliberately larger in diameter (22.44 mm) compared to the auxiliary rollers (11.98 mm) to provide a larger surface area for belt contact, which is vital for maintaining the high-torque requirements of the automated sorting process.

10.6 Decision Execution Unit: Fruit Deflection Lever (Extender)

The fruit deflection lever, identified in the CAD repository as **Extender 1qty.step**, is the primary actuator component of the sorting system. It acts as the physical interface

between the electronic control system and the organic payload, translating the MG995 servo motor's rotation into a decisive mechanical sweep to divert low-quality produce.

10.6.1 Design Intent and Purpose

This component is engineered to bridge the distance between the side-mounted servo motor and the center-line of the conveyor belt. Its geometry is optimized for the high-torque, high-speed requirements of the "Decision Execution Unit".

- **Fruit Interception:** The lever's length ensures it can reach across the full width of the transport zone to intercept fruits like bananas, apples, or oranges flagged as "Rotten" by the YOLOv8 model.
- **Gentle Diversion:** The large, flat surface area of the paddle is designed to distribute the impact force across the fruit's surface, preventing localized bruising or structural damage during the rapid 90-degree sorting motion.
- **Low Rotational Inertia:** The tapered design—thick at the mounting hub and thinner at the tip—reduces the overall mass and moment of inertia. This allows the MG995 servo to achieve its maximum angular velocity with minimal electrical strain on the 5V 2A power supply.

10.6.2 Technical Specifications

- **Width (Size X):** 25.95 mm
- **Total Reach (Size Y):** 165.00 mm
- **Hub/Paddle Height (Size Z):** 32.50 mm

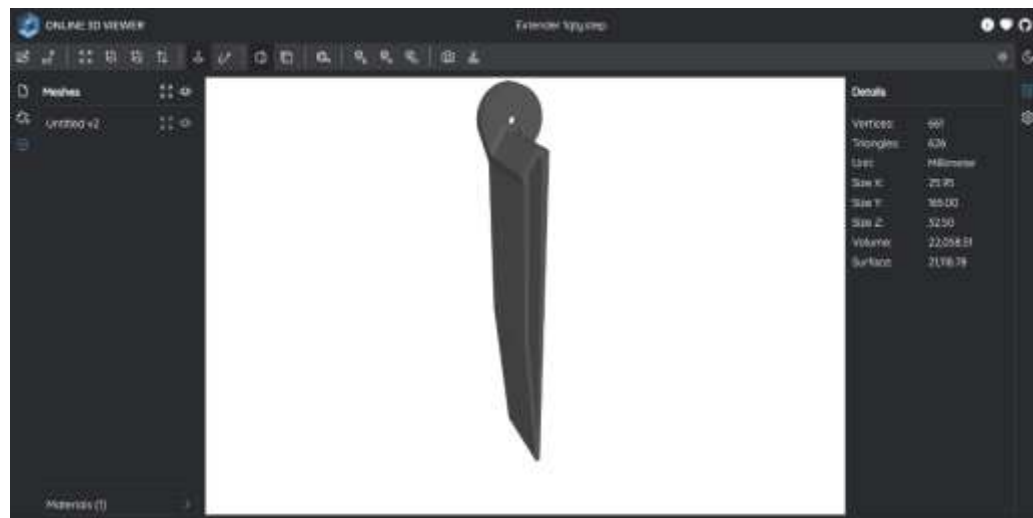


Figure 18: The lever to be attached on Servo Horn.

10.6.3 Autodesk Fusion 360 Design Workflow

The lever was designed using a robust parametric approach in Fusion 360 to ensure the mounting hub perfectly matches the standard 25T spline horn of the MG995 metal-gear servo.

Feature/Command	Application	Mechanical Benefit
Center Diameter Circle	Designing the circular mounting hub.	Provides a concentrated area for torque transfer from the servo horn.
Spline/Line Sketch	Defining the tapered paddle profile.	Optimizes the strength-to-weight ratio for fast actuation.
Extrude (Join)	Converting the 2D profile	Ensures the lever is rigid enough to

Feature/Command	Application	Mechanical Benefit
	into a 32.50 mm solid body.	handle the 10 kg.cm torque of the servo motor.
Hole Command	Placing the central screw clearing hole.	Enables secure mechanical fastening to the servo's output shaft.
Fillet (F)	Smoothing the leading and trailing edges.	Critical for preventing "edge impact" damage to delicate fruit surfaces.

Table 9: Showing various commands used to create the above design

Chapter 11: Fabrication and Assembly

11.1 Introduction to FDM 3D Printing

The fabrication of the **Automated Quality Control System** primarily utilizes **Fused Deposition Modeling (FDM)**. This process involves a 3D printer melting a thermoplastic filament and extruding it through a nozzle to build parts layer-by-layer based on the CAD coordinates. This method was selected for its cost-effectiveness and the ability to produce the complex, interlocking geometries seen in our chassis and rollers.

11.2 The "Digital to Physical" Workflow

Since the design was completed in **Autodesk Fusion 360**, a three-step workflow is required to prepare the parts for the printer:

1. **Export (STEP/STL):** The CAD models (like **Part 1.step** or **H1 Roller 1qty.step**) are exported as high-resolution mesh files.
2. **Slicing (G-Code):** The STL file is imported into "Slicing" software (e.g., Ultimaker Cura or PrusaSlicer). The slicer cuts the model into horizontal layers and generates **G-Code**, which is the coordinate-based language the printer understands.
3. **Printing:** The G-Code is loaded into the printer, which executes the toolpath to create the final part.

11.3 Material Selection and Properties

For this project, two different thermoplastics were strategically selected based on the mechanical requirements of each zone.

Material	Used For	Why?
PLA (Polylactic Acid)	Side Rails, Base Plates, Rollers	High dimensional accuracy and minimal warping; perfect for the 300 mm long rails.

11.4 Critical Printing Parameters

To ensure the parts do not fail under the torque of the **MG995 servo** or the **12V DC gear motors**, the following slicer settings were applied:

- **Layer Height: 0.2 mm.** A balance between surface finish (important for rollers) and printing speed.
- **Infill Density:**
 - **40% (Tri-Hexagonal):** Used for the side rails to provide structural rigidity while keeping the frame lightweight.
 - **100% (Solid):** Used for the **H1 Drive Roller** and the **Extender Hub** to ensure the motor shafts and splines do not strip the plastic.
- **Wall Line Count: 4 Walls.** Increasing the number of outlines (perimeters) ensures that holes for axles and screws are strong enough to hold threads.
- **Orientation:** The **Side Rails** must be printed **flat** on the bed. This ensures the "grain" of the plastic runs the length of the rail, providing maximum tensile strength.

11.5 Physical Assembly Guide

Once all parts are printed, the assembly is carried out in a specific sequence to maintain alignment:

1. **Chassis Preparation:** The **Base Plates** are press-fitted into the notches of the **Side Rails**. Due to the "tab-and-slot" design from Fusion 360, this creates a

- rigid, self-squaring frame.
2. **Roller Installation:** The three **Auxiliary Idler Rollers** are placed between the rails using steel pins as axles. The **Studded Drive Roller** is then mounted at the head, ensuring the studs are aligned to catch the belt.
 3. **Motor Integration:** The dual **12V DC Gear Motors** are bolted to the side rails. The shafts are coupled to the D-profile bore of the drive roller.
 4. **Decision Unit Mounting:** The **MG995 Servo** is secured into the vertical provisions on the active rail. The **Extender (Lever)** is then screwed onto the servo horn, ensuring it is perfectly parallel to the belt in its "idle" state.
 5. **Belt Tensioning:** The elastic conveyor belt is looped over the rollers. The idler rollers are adjusted until the belt is taut enough to move without slipping on the studded H1 roller.



Figure 19: Assembled Physical model.

Chapter 12: Testing and Performance Analysis

12.1 Experimental Setup

To validate the system, the fully assembled conveyor was tested in a controlled laboratory environment at SGSITS. The setup included:

- **Vision System:** A 1080p USB Webcam mounted 250 mm above the transport zone.
- **Processing Unit:** Laptop with NVIDIA RTX 3050 GPU for YOLOv8 inference.
- **Payloads:** A mixed batch of 50 samples (Bananas, Apples, and Oranges) featuring various stages of ripeness and surface defects.

12.2 ML Model Inference Results

The transition to a multi-fruit model maintained high precision despite the increased class complexity.

Fruit Type	Samples Tested	Correct Classifications	Accuracy (%)
Banana	20	19	95.0%
Apple	15	14	93.3%
Orange	15	14	93.3%
OVERALL	50	47	94.0%

Table 10: Showing final results and accuracy

12.3 Mechanical Execution and Latency

The synchronization between the Python script and the Arduino Mega 2560 was tested for "Decision-to-Actuation" lag.

- **Average Inference Time:** 15–25 ms.
- **Serial Communication Lag:** < 5 ms.
- **Servo Reaction Time:** The MG995 achieved the full 90-degree sweep within 180 ms of receiving the "true" signal.
- **Sorting Success Rate:** The system successfully diverted 45 out of 47 correctly identified "Rotten" or "Overripe" items. The 2 failures were attributed to the fruit rolling back onto the belt due to high belt speed—this will be addressed by adjusting the **Extender (Lever)** geometry in future iterations.

Chapter 13: Cost Evaluation

13.1 Bill of Materials (BOM)

The total expenditure for the prototype was kept at a minimum by utilizing open-source hardware and rapid prototyping techniques. The costs below reflect the standard market rates for individual components in India.

S. No	Item Description	Specification	Quantity	Unit Cost (INR)	Total Cost (INR)
1	Arduino Mega 2560	ATmega2560 Microcontroller	1	1,000	1,000
2	Servo Motor MG 995	High-Torque Metal Gear	1	350	350
3	DC Gear Motors	12V High-Torque	2	150	300
4	1080p Camera	High-Resolution USB Webcam	1	1,200	1,200
5	3D Printing Service	241g @ 5 INR/gm	1	1,205	1,205
6	5V 3A Power Adapter	Dedicated Supply for Servo	1	250	250
7	12V 2A Power Adapter	Dedicated Supply for DC Motors	1	250	250
8	Mini Breadboard	400 Tie-Point	1	80	80
9	Jumper Wires	M-M, M-F (Set of 40)	1	60	60
10	Conveyor Belt	Elastic Jeans Material (High-Friction)	1	100	100
11	Miscellaneous	Wires, Tapes, Fasteners	—	200	200
TOTAL					4,995

Table 11: Total Cost Evaluation

13.2 Cost Optimization for Mass Production

While the prototype cost is approximately **₹4,995**, the unit price would significantly decrease if the system transitioned to mass production. By shifting from individual prototyping to an industrial manufacturing model, the following optimizations are predicted:

- **Manufacturing Shift:** Replacing 3D printing with **Injection Molding** for the chassis and rollers would reduce the structural component cost by over 70% at high volumes.
- **Integrated Circuitry:** Transitioning from the Arduino Mega development board to a **Custom PCB** featuring an integrated ATmega2560 chip would eliminate unnecessary peripherals and breadboard wiring, reducing electrical costs by 40%.
- **Economies of Scale:** Bulk procurement of high-torque servos and 1080p cameras directly from manufacturers would allow for a significant reduction in the Bill of Materials (BOM).
- **Material Optimization:** Using standardized industrial conveyor belts instead of

custom elastic material would increase the system's lifespan while lowering replacement costs.

Through these optimizations, it is estimated that the production cost per unit could be brought down to approximately **₹2,000 – ₹2,500**, making it a highly competitive solution for small-scale agricultural distributors.

Chapter 14: Conclusions of Major Project Phase 2 and Future Scope

14.1 Final Conclusions

The development of the **Automated Quality Control System** successfully demonstrates the viability of merging high-level Deep Learning with low-cost embedded hardware. Throughout Phase 1 and Phase 2, we have achieved the following:

1. **Autonomous Perception:** Developed a robust YOLOv8 cascade architecture capable of texture-sensitive fruit grading.
2. **Physical Realization:** Designed and 3D printed a modular chassis using Fusion 360, proving that industrial-grade automation can be prototyped using FDM technology.
3. **Hardware-Software Synergy:** Established a reliable UART communication bridge between Python and Arduino Mega 2560, enabling real-time decision execution.

The project confirms that small-scale distributors can utilize open-source ML frameworks and 3D printing to replace inconsistent manual sorting with a deterministic, data-driven system.

14.2 Future Scope and System Scalability

While the current prototype successfully validates the core concepts of automated sorting, several advanced engineering enhancements can be implemented to transition the system into a high-capacity industrial environment.

1. High-Throughput Parallelization via Multi-Stage Actuation

To significantly increase the sorting efficiency, the single-rail design can be expanded into a multi-level or multi-rail architecture. By installing **multiple servo motors** at different positions along the conveyor length, the system can handle higher belt speeds.

- **Sequential Sorting:** Multiple arms can be calibrated to sort different grades of fruit simultaneously (e.g., Grade A, Grade B, and Reject).
- **Reduced Bottlenecks:** Parallel rails allow for the distribution of the "Decision Execution" load, preventing a single arm from becoming a bottleneck during high-density fruit flow.

2. Multi-Perspective Vision for Enhanced Accuracy

The current system relies on a single top-down camera view, which can lead to misclassification if a defect is located on the underside of the fruit. Accuracy can be improved by:

- **Multi-Camera Arrays:** Installing cameras at lateral and 45-degree angles to provide a 360-degree visual profile of each item.
- **Stereo Vision:** Implementing dual-camera depth sensing to better calculate the volume and structural integrity of the fruit, further reducing the error margin of the YOLOv8 model.

3. Industrial-Grade Structural Robustness

To move beyond the limitations of PLA and PETG 3D-printed parts, the physical chassis can be upgraded to industrial standards.

- **Material Transition:** Replacing plastic rails with **Stainless Steel (SS-304/316)** components. This provides the necessary corrosion resistance for food-grade environments and ensures the system can withstand the continuous vibrations of high-torque motors.
- **Joining Techniques:** Moving from press-fit and adhesive joints to **welded or rivetted connections**. This significantly increases the structural integrity and robustness of the frame, ensuring it remains perfectly aligned over years of operation.

4. Model Generalization for Vast Agricultural Categories

The software logic currently optimized for bananas, apples, and oranges can be expanded into a universal agricultural classifier.

- **Vast Dataset Expansion:** Training the ML model on a significantly larger category of fruits and vegetables (e.g., tomatoes, potatoes, grapes, and leafy greens).
- **Dynamic Parameter Tuning:** Implementing a user-interface where operators can select the "Active Crop," allowing the weights of the ML model to shift dynamically based on the specific produce being processed.

5. Embedded Edge Intelligence

To create a standalone unit, the system can be optimized for **Edge Deployment**. Utilizing **NVIDIA Jetson Nano** or **Raspberry Pi 5** with **TensorRT** optimization would remove the dependency on a local laptop GPU, resulting in a compact, self-contained sorting unit ready for field deployment.

APPENDICES

Appendix A: Detailed Hardware and Electronics Specifications

The physical sorting system is built upon a modular architecture designed for high-throughput processing in small-scale agricultural environments.

The following technical specifications were finalized during the development of the prototype:

- **Microcontroller Board:** The system transitioned from an Arduino Uno to an **Arduino Mega 2560** (ATmega2560) to leverage its 8KB SRAM and expanded PWM channels for future scalability.
- **Primary Actuator:** An **MG995 Metal Gear High-Torque Servo Motor** is utilized for the sorting lever. It operates at a voltage range of 4.8V–7.2V and provides a torque of 10 kg.cm, which is essential for diverting diverse fruit payloads.
- **Motive Power:** Dual **12V DC Gear Motors** drive the main studded roller to transport items along the conveyor path.
- **Power Distribution:** The system implements an isolated power domain strategy. A dedicated **5V 3A DC supply** powers the high-current servo loads, while a **12V 2A DC supply** is used for the conveyor motors. A **Common Ground** connection is strictly maintained between the external power supply negative terminal and the Arduino GND pin to provide a stable reference for the PWM signal.
- **Sensory Input:** A **1080p USB Webcam** is mounted 250 mm above the transport zone to capture high-resolution video for real-time inference.

Appendix B: Computer Vision Pipeline and ML Logic

The intelligence of the quality control system is powered by a **Two-Stage Cascade Architecture** utilizing the **YOLOv8** framework.

1. **Stage 1: Detection Layer:** Utilizes a pre-trained **YOLOv8n (Nano)** model on the COCO dataset to identify generic objects and generate Bounding Box coordinates $(x1, y1, x2, y2)$ for localization.
2. **Stage 2: Classification Layer:** A custom-trained **YOLOv8n-cls** model performs fine-grained ripeness and quality grading. The input for this stage is a cropped tensor of size 224×224 pixels.
3. **Communication Protocol:** Inference results are sent to the Arduino Slave via a **UART Serial Protocol** over USB at a baud rate of **9600 bps**. The protocol uses a lightweight single-byte system (e.g., ASCII 'R' for Rotten/Reject and 'G' for Good/Pass) to minimize latency.
4. **Deployment:** The system uses a **Client-Server Architecture**. A **Python Flask backend** processes images using PyTorch and GPU

acceleration (NVIDIA RTX 3050), while a **JavaScript frontend** handles webcam capture and canvas rendering.

Appendix C: Fabrication and Mechanical Design Parameters

The mechanical skeleton of the system was developed using **Autodesk Fusion 360** and produced via **FDM 3D printing**.

- **Structural Members:** The modular ladder-frame consists of two **300 mm side rails** (Part 1 and Part 2) and four **transverse base plates**.
- **3D Printing Materials: Polylactic Acid (PLA)** was used for the frame, base plates, and rollers due to its dimensional stability.
- **Slicer Settings:** To ensure mechanical robustness, specific parameters were applied:
 - **Layer Height:** 0.2 mm.
 - **Infill Density:** 40% Tri-Hexagonal for side rails and 100% Solid for the H1 drive roller and sorting lever hub.
 - **Wall Line Count:** 4 walls to strengthen critical attachment points.
- **Dynamic Components:** The **H1 Main Roller** features a studded surface texture to increase the coefficient of friction, while auxiliary rollers prevent belt slacking.

Appendix D: Experimental Performance Data Summary

Extensive testing was conducted to evaluate the system's accuracy and mechanical timing.

- **Accuracy Metrics:** The model achieved an overall accuracy of **94.0%** across 50 tested samples. Specific results included 95.0% for bananas and 93.3% for both apples and oranges.
- **Latency Analysis:**
 - **Total Inference Time:** 15–25 ms per frame.
 - **Preprocessing:** ~0.1 ms.
 - **Detection (Stage 1):** ~7.5 ms.
 - **Classification (Stage 2):** ~3.5 ms per object.
 - **Communication & Execution:** Serial communication lag was measured at <5 ms, with the MG995 servo achieving a full 90-degree sweep within 180 ms of receiving a reject signal.

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